Beyond Geography: Predicting and Preventing Food Safety Failures in Boston Establishments

Sarah Yeh '27 Data Science Major Capstone

Introduction

Research Question

To what extent do geographical areas (Census tracts) predict the frequency of total health violations in Boston food establishments compared to restaurant-level operational behaviors, and how can predictive modeling be used to prevent systemic health code failures?

Background

Observation suggests that health code violations are disproportionately clustered in certain Boston census tracts, making it critical to investigate whether these disparities are driven by geographical factors or restaurant-level operational behaviors.

Data

Sources

- Boston Food Inspection Data (2006-2026)
- US Census Tract Data (TIGER/Line Shapefiles and corresponding socioeconomic data (American Community Survey)

Data Processing

- **Filters**: Restricted to stationary establishments and violation levels *, **, *** . Mobile units **excluded**.
- **Data Cleaning**: Conducted removal of observation rows containing missing location data (address, coordinates, or ZIP) (approx. 74,000 removed out of 400,000 total obs.).
- **Feature Engineering**: Calculated the inspection_gap (days since visit) and generated operational flags (pests, debris, hygiene, chain status).
- **Merging**: Geocoded locations for an accurate spatial merge at the Census Tract level; added **median_income** and **pct_poverty**.
- **Unit of analysis**: Each row represents a restaurant per year.

Visualizations

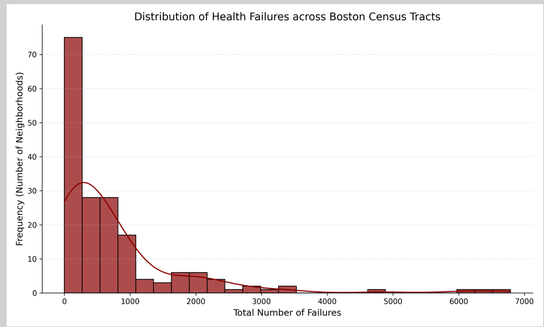

Figure 1. *Distribution of health failures across Boston Census Tracts.*

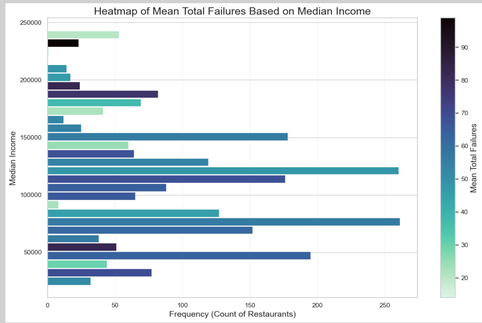

Figure 2. *Initial Mapping of Mean Failures by Median Income.*

Figure 3. *Graph showing the frequency of violation types by neighborhood income level, showing that there are more pest violations in lower income neighborhoods.*

Figure 4. *Impact of Violation Flags on Total Violation Count.*

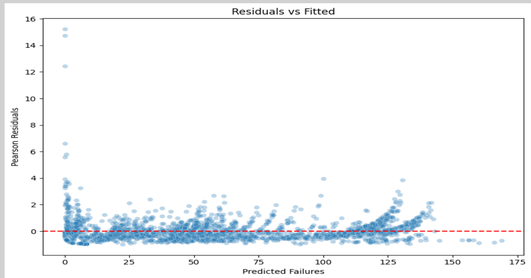

Figure 5. *Residual vs. Fitted Plot of the Negative Binomial Model verifying the model's assumptions.*

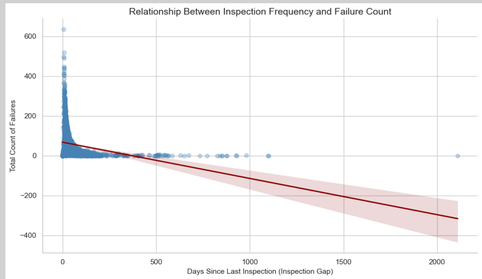

Figure 6. *Relationship between Inspection Frequency and Failure Count showing that the more a restaurant fails, the more that they will have inspections.*

Conclusion

- **Management > Geography:** While raw counts suggest geographical clustering, modeling reveals that operational behaviors are true drivers of risk. Neighborhood income was statistically insignificant (p=0.68).
- The Negative Binomial model explained **54.61%** of the variance, proving food safety risk is highly predictable when focusing on internal kitchen culture rather than geographical factors.
- **Proactive Action:** Current patterns confirm a "reactive" strategy (shorter inspection gaps for repeat offenders). This model provides a framework for Boston to aim towards **proactive** risk-tiering based on behavioral flags.

Methods/Results

Method

- Data Integration: Merged 20 years of Boston Food Inspection data with Census Tract shapefiles using a spatial join.
- Model Specification: Utilized a **Negative Binomial Generalized Linear Model**.
- Multicollinearity: Conducted **VIF** analysis; all factors remained below 2.5, confirming no significant multicollinearity.
- Compared base GLM model (without neighborhood level variables) and full GLM model (including neighborhood variables)
- Train/Test split: Model was trained on data from 2006-2024 (2218 obs.) and validated using modern data from 2025-2026 (138 obs.).

Model	AIC	Pseudo R2
Base Model	**20519.14**	0.7469
Full Model	20523.64	0.7471

Results

- After comparing base model and full model using the **Likelihood Ratio Test** (LRT) with (**p = 0.68**) and **AIC** analysis, it is revealed that neighborhood demographics **do not** provide predictive power beyond operational flags.
- Used **base model** to predict total_failures in restaurants.
- Train data resulted in a Pseudo R2 of **0.7133**.
- Predictive Power: Test data resulted in a Pseudo R2 of **0.5461**

Variable	Coeff (log)	P-value
Flag_hygiene	1.1820	< 0.001
Flag_pests	0.7741	< 0.001
Flag_debris	1.2928	< 0.001
Is_chain	-0.1494	0.385
inspection_gap	-0.0072	< 0.001
year	-0.0924	< 0.001

Metric	Training Set (2006-2024)	Test Set (2025-2026)
Pseudo R2	0.7133	0.5461
Mean Absolute Error (MAE)	28.02	5.80
RMSE	49.85	9.13

Limitations

- **Surveillance and Reporting Bias:** Lower-income areas may face disproportionate inspection frequency.
- **Lack of Variables:** The dataset lacks internal establishment metrics such as staff turnover rates or management experience.
- **Data Loss**: Some data was loss through the spatial join, which may have excluded some Census tracts.